SURVEY RESEARCH PRACTICE

QUESTION PHRASING, WORDING AND DESIGN

INTRODUCTION TO SURVEY RESEARCH

- **Briefly** explain the purpose and significance of survey research
- Discuss strengths and weaknesses of survey research
- **Highlight** the use of survey research in understanding human behavior, attitudes, and preferences
- Review problems with survey research that cause systematic errors

SURVEY DESIGN PROCESS

- **Sampling**:
  - Recall the importance of random selection- selection of participants using a random sampling method (not random assignment)
    - EPSEM sampling like random sampling, stratified sampling, and convenience sampling
    - Purpose is to obtain a sample that represents a population
  - If you use an EPSEM techniques, the resulting random sample will be similar to the population
    - **Example:** Randomly select 1,000 people from the adult population in Columbus, Ohio, your sample would be similar to the adult population of Columbus, Ohio (and may or may not be similar to the adult population of East Lansing, MI).
  - Very important for survey research in which you need to generalize from a single sample directly to a population
  - Example: random_numbers.R

PROS AND CONS

- Strengths of survey research
  - Particularly useful in describing large populations (if appropriate sampling was used)
  - Standardized questionnaires can ensure uniform responses and measurement
  - Some protection against respondents interpreting concepts differently

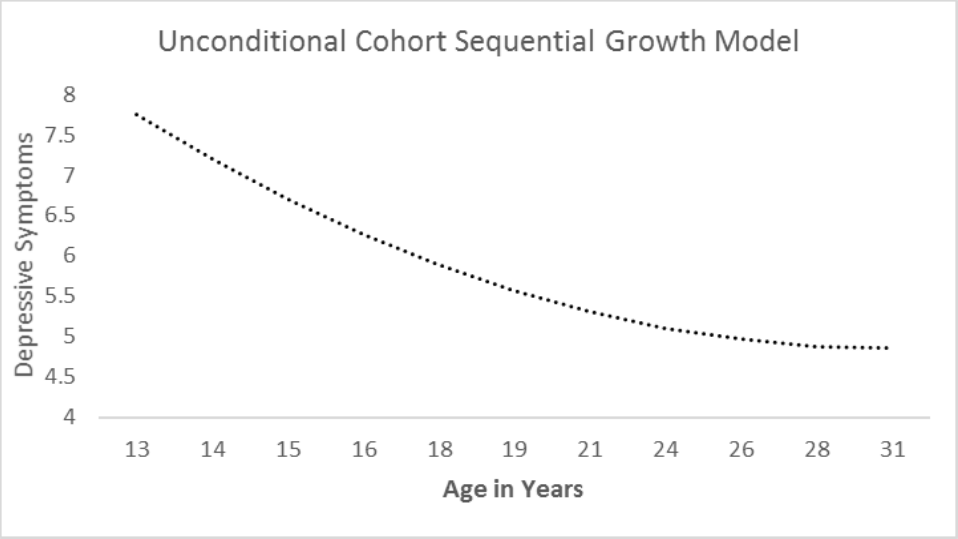
- Weaknesses
  - Standardized questionnaire items often represent the least common denominator in assessing people's attitudes, orientations, circumstances, and experiences
  - Superficial coverage of complex topics
  - Survey research cannot readily deal with the specific contexts of social life
  - Some populations might be hard to contact through customary sampling methods

SELF-REPORTS

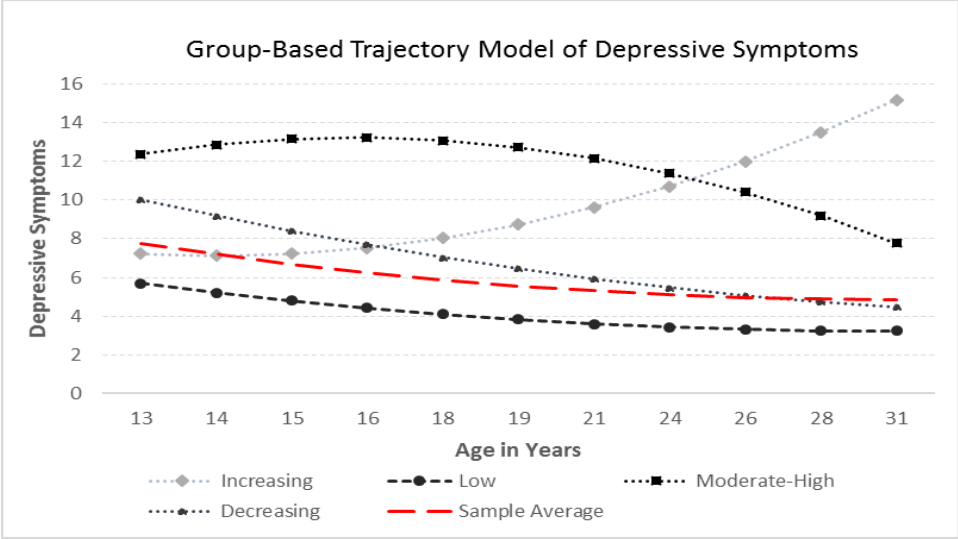
- Asking questions is the best available method to explore or explain why people ***behave*** in certain ways, endorse certain ***attitudes*** and/or have acquired certain "***conditions***"
- Within the category of "self-report" two different applications are distinguished by the sampling method
  - Non-random; and
    - Example: Studies of clinically depressed people are often clinical samples
  - Random
    - Studies of the general population select random samples and the focus is on depression prevalence
  - It is important to distinguish between clinical samples and random samples
    - Example: ADD health

EXAMPLE: SELF-REPORT DATA FROM THE ADD HEALTH SURVEY

Self-reported depressive symptoms, maltreatment and delinquent behavior



Youth who engaged in delinquent behavior, and/or who reported physical and sexual abuse are more likely to have an initially high but decreasing depressive symptom trajectories



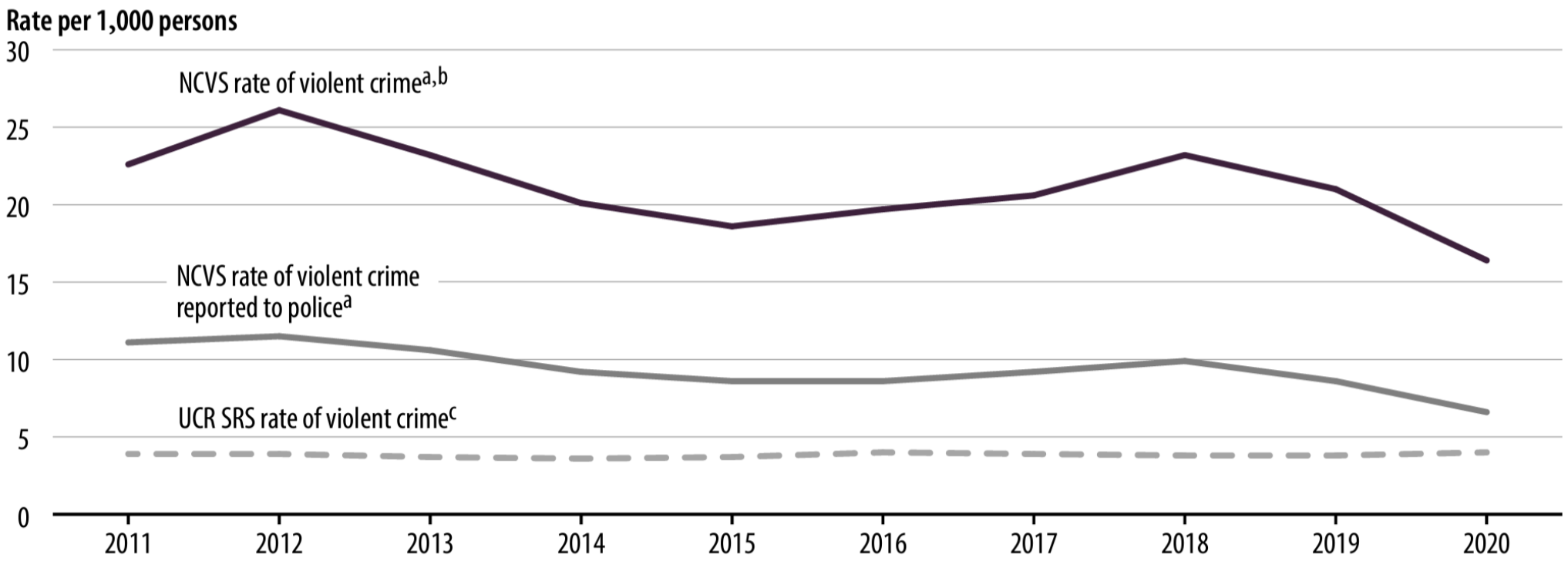
TARGETED SURVEYS FOR EVALUATION

- Sometimes a big representative sample is not effective for the question at hand
- Suppose I want to evaluate a program aimed at minimizing youth delinquent behaviors
- Best thing to do is an evaluation study that uses pre/post testing and a quasi-experimental design

WHEN SURVEY RESEARCH IS PARTICULARLY GOOD

- Surveys are great to gauge perceptions, behaviors and attitudes
- Example: violent crime victimization
  - The National Crime Victimization Survey versus Uniform Crime Reports
  - Trends suggest huge differences in rate of violent victimization

EXAMPLE: COMPARE SELF-REPORT TO AGENCY DATA



https://bjs.ojp.gov/content/pub/pdf/ntcm1120.pdf

HEURISTICS & BIASES

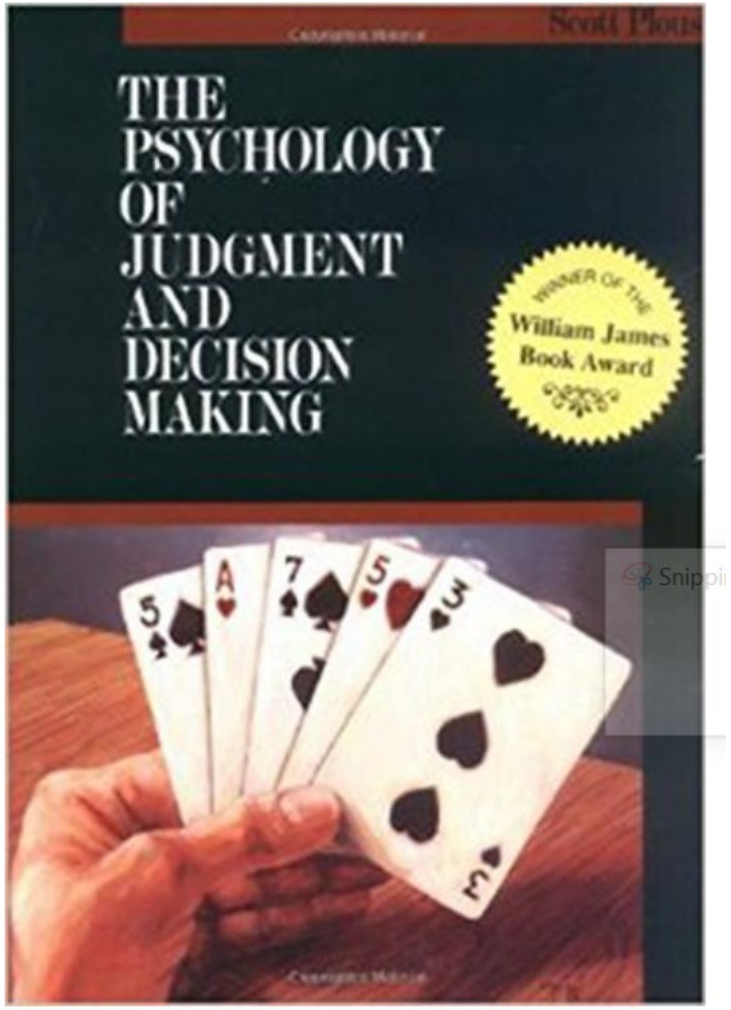
- How do they affect people's judgments and decisions on surveys
- How do they affect researchers and/or research

GET A CUE!

- "…people rely on a limited number of heuristic principles which reduce complex tasks of assessing probabilities and predicting values to simpler judgmental operations. In general, these heuristics are quite useful, but sometimes they lead to severe and ***systematic*** errors (Tversky & Kahneman, *Science,* 185, p. 1124-1131).

PERCEPTION, MEMORY AND CONTEXT CUES

- People see what they expect to see
- People reconstruct memories
- People try to be "consistent"
- More recent and more distant things affect perception
- People believe good things about good … (people, places, and things)



DIFFICULTIES WITH SELF-REPORT DATA

- General population surveys and surveys of specific populations may result in different types of systematic errors
  - Sampling/selection bias
  - Recall bias
  - Social desirability
  - Fabricated response bias
  - Reluctance to disclose

SELECTION & SAMPLING BIAS

- **Selection Bias**
  - **Description**: Occurs when the sample of participants is **not representative of the entire population**
  - **Impact**: If certain groups are overrepresented or underrepresented, the findings may not generalize accurately
  - **Mitigation**: Use random sampling techniques and ensure diverse participation
- **Sampling Bias**
  - **Description**: Results from an unrepresentative sample due to **flaws in the sampling process**
  - **Impact**: same as above
  - **Mitigation**: same as above

RECALL BIAS

- **Description**: Participants may inaccurately recall past events or experiences
- **Impact**: Responses based on faulty memory can lead to unreliable data
- **Mitigation**: Use prompts, time frames, and memory aids to improve recall accuracy
- **Note**: recall bias is itself biased….

RESPONSE BIAS

- **Description**: Participants may provide inaccurate or socially desirable responses; Participants tend to answer in a way that aligns with societal norms or what they perceive as socially acceptable especially during face-to-face interviewing
- **Impact**: Data may be skewed due to participants' desire to conform or present themselves favorably; Responses may not reflect true feelings or behaviors
- **Mitigation**: Anonymity, neutral wording, and minimizing leading questions can reduce response bias

*The Direction of Race of Interviewer Effects Among African-Americans: Donning the Black Mask**

Darren W. Davis, *Michigan State University*

Theory: The discussion that takes place between African-American respondents and interviewers of different races is symbolic of the normal everyday sensitivity to race and strangers of different races. Race of interviewer effects can usefully serve as an indicator of evolving areas of interpersonal tension between African-Americans and whites, and deserve to be treated as a fact of social life and not merely as an artifact of the survey interview. Underlying the potential response bias and estimation problems created by the race of interviewer effects rests a set of coherent beliefs about race and perceived constraints on the freedom of expression.
Hypothesis: Given the persistence of negative reactions to African-Americans among whites, African-Americans can be expected to be more sensitive to white interviewers. Characteristic of the traditional role-playing behavior meant to appease and accommodate whites, African-Americans out of a sense of fear or intimidation are expected to conceal their true political beliefs and place self-imposed limits on their freedom of expression in response to the white interviewer.
Methods: Ordinary Least Squares analysis is performed on survey data from the 1984 National Black Election Study.
Results: As a sign of deference to the interviewer, African-Americans in response to white interviewers are more likely to acquiesce to mutually contradictory evaluations of both the Democratic and Republican parties, to both Ronald Reagan and Jesse Jackson, and to black officials supportive of both Ronald Reagan and Jesse Jackson. More importantly, alterations in the interviewers' race from the preelection to postelection panel waves reveal that when the change is from an African-American to a different white interviewer, or the same white interviewer, African-Americans are more likely to admit that blacks do not have the power to change things, that blacks cannot make a difference in presidential and local elections, that blacks should not form their own political party, and that whites do not keep blacks down.

NON-RESPONSE BIAS

- **Description**: Occurs when certain participants refuse to participate or drop out, which happens when participants are followed over time
- **Impact**: Non-response can introduce bias if those who participate differ systematically from non-respondents
- **Mitigation**: Analyze non-response patterns and adjust weights if necessary

Variable	n	Weighted Proportion	Proportion Among Those With Missing Data
Gender			
Male	3255	0.34	0.35
Female	7854	0.66	0.65
Age, yrs			
18–40	4734	0.42	0.29
41–50	2329	0.21	0.19
51–60	1772	0.16	0.16
61–70	1117	0.10	0.13
71–80	847	0.07	0.14
>81	310	0.03	0.09
Race/ethnicity			
White	6637	0.72	0.62
Black	1675	0.12	0.16
Hispanic	2141	0.11	0.18
Other	656	0.06	0.05
Education			
Less than high school	2504	0.15	0.33
High school graduate	3519	0.32	0.35
Some college	5086	0.53	0.32
Diabetes			
No	10,186	0.93	0.89
Yes	923	0.07	0.11

RESULTS

Table 1 shows sociodemographic and clinical characteristics, in terms of unweighted counts (n = 11,109) and proportions weighted to represent the US civilian noninstitutionalized population. Table 1 also shows analogous proportions among the 1179 respondents removed from the analytic sample due to missing data. Respondents with missing data were more likely to be older, to be black or Hispanic, to be less educated, to have a chronic condition (other than asthma), to have more pain, and to have worse self-reported health and self-reported mental health.

Fleishman, J. A., & Zuvekas, S. H. (2007). Global self-rated mental health: associations with other mental health measures and with role functioning. *Medical Care*, *45*(7), 602-609.

FABRICATED RESPONSE BIAS

- **Description**: Occurs when certain participants are not truthful, maybe because of social desireability or maybe its just none of your business...
- **Impact**: Fabricated responses impact the validity of the study
- **Mitigation**: Question wording can reduce the bias; multiple question on same topic
- **Example**: How would you get people to admit to deviant behavior or accurately report their earnings?
  - There are a lot of things which are actually crimes, but which are done by lots of people, and which many people do not think of as crimes. On this card [card handed to respondents] are a list of 8 items of them. For each one, can you tell me how many people you think do it –
    - most people,
    - a lot of people, or
    - no one.
  - Take office supplies from work when not supposed to…
  - Next, Rs were asked if *they* ever did X…
- Why might this minimize the tendency to be untruthful?

ORDER BIAS

- **Description**: The order in which questions are presented influences responses
- **Impact**: Early questions can prime participants for later ones, affecting their answers
- **Mitigation**: Randomize question order or use counterbalancing
- More below

TYPES OF SURVEYS

- **Questionnaire Surveys**:
  - Use of standardized questionnaires
  - Data is collected through structured questions
  - Data is collected in various formats (postal, group-administered, online)
- **Interview Surveys**:
  - Can be F2F, telephone, and focus group interviews
  - Advantages and limitations

QUESTIONNAIRE DESIGN

- Emphasize clarity, validity, and reliability
- Address potential biases discussed above

GUIDELINES FOR ASKING QUESTIONS

- How questions are asked is the single most important feature of survey design
- Research concepts are operationalized by asking questions
- We need to understand how question wording effects answers… Here are the most important
  - Framing and other Psychological effects
  - Order of questions
  - Context in which the question appears
  - Format
  - Filters and contingencies
  - Phraseology

QUESTIONS AND STATEMENTS

- Researchers must summarize the attitude or behavior in a fairly brief statement then present the statement and ask for agreement or disagreement
- Problems
  - If someone holds an attitude or behavior we expect that their actions will reflect that attitude or behavior
  - Attitude-attitude inconsistency
    - Americans tend to support equality but oppose racial preferences
  - Attitude-behavior inconsistency
    - People love to say they love to exercise but engage in no physical activity
  - Psuedo-opinions (below)

AUGUST 8, 2023 2:20PM

Americans Say they Like "Affirmative Action" Yet Oppose Racial Preferences in College Admissions

CLASSIC STUDIES IN SURVEY RESEARCH & QUESTION WORDING

- The way a question is worded or framed may affect individuals' responses to that question
  - "safe" versus "safer"
    - 40% of the respondents of a poll in Britain said the nuclear weapons of their country made them "feel safe."
    - When the question was asked slightly differently, 50% of the respondent said the nuclear weapons of their country made them "feel safer."
  - "long" versus "short"
    - How long was the move? 130 min. How short was the movie 100 min.
  - "faster" or "fast"
    - "Should the U.S. Army withdraw from Vietnam faster or slower?" – 42% "faster," 16% "slower," 29% "same as now."
    - "Is the pace with which the U.S. Army withdraws from Vietnam too fast, too slow, or about right?" – 29% "too slow," 6% "too fast," 49% "about right."

FRAMING, PHRASING AND PRESENTATION

- Question phrasing, answer choices, and question presentation affects responses!

Framing

On September 27, 2001 President Bush put forth the Hostile Powers Resolution **forbidding** UN Humanitarian Economic aid to Afghanistan. Was president Bush within his powers as President to enact such a UN resolution?

A) Yes, Bush was definitely within his power.

B) No, Bus overstepped his authority.

Ordering

vs. not allowing?

FRAMING, RISK AVERSION AND RISK SEEKING

- The Certainty Effect: people tend to overweight outcomes that are perceived as ***certain*** or ***highly probable***, while underweighting outcomes with lower probabilities
- Imagine a patient diagnosed with a medical condition. They have two treatment options:
  - **Option A**: A treatment with a **100% success rate** but some side effects.
  - **Option B**: A treatment with a **90% success rate** but a **10% chance of no side effects**.
- When faced with a choice between a sure gain and a risky gain, the certainty effect causes individuals to prefer the sure gain.
- Similarly, when comparing a sure loss and a risky loss, people tend to avoid the sure loss, even if the expected values are the same

PLASTICITY AND FRAMING

- In some cases, offering the same choice in two different contexts can lead to very different answers
- For example, suppose you were faced with the following choice:
  - Alternative A: A 100 percent chance of losing $50
  - Alternative B A 25 percent chance of losing $200, and a 75 percent chance of losing nothing

  Which alternative would you choose?
- 80 percent of the people asked by Paul Slovic, Baruch Fischhoff, and Sarah Lichtenstein (1982) preferred Alternative B
- Most people are "risk seeking" when it comes to losses; that is, they prefer to risk a relatively large loss (in this case, $200) rather than suffer a sure loss with the same expected value (?)

Risk preferences and aging: The “Certainty Effect” in older adults’ decision making

Mara Mather, Nina Mazar, Marissa A. Gorlick, Nichole R. Lighthall, Jessica Burgeno, Andrej Schoeke, and Dan Ariely

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The publisher's final edited version of this article is available at Psychol Aging

Abstract

Go to:

A prevalent stereotype is that people become less risk taking and more cautious as they get older. However, in laboratory studies, findings are mixed and often reveal no age differences. In the current series of experiments, we examined whether age differences in risk seeking are more likely to emerge when choices include a certain option (a sure gain or a sure loss). In four experiments, we found that age differences in risk preferences only emerged when participants were offered a choice between a risky and a certain gamble but not when offered two risky gambles. In particular, Experiments 1 and 2 included only gambles about potential gains. Here, compared with younger adults, older adults preferred a certain gain over a chance to win a larger gain and thus, exhibited more risk aversion in the domain of gains. But in Experiments 3 and 4, when offered the chance to take a small sure loss rather than risking a larger loss, older adults exhibited more risk seeking in the domain of losses than younger adults. Both their greater preference for sure gains and greater avoidance of sure losses suggest that older adults weigh certainty more heavily than younger adults. Experiment 4 also indicates that older adults focus more on positive emotions than younger adults do when considering their options and that this emotional shift can at least partially account for age differences in how much people are swayed by certainty in their choices.

PHRASES AND WORDING

- The meaning of a given response depends on the wording and the context
  - Avoid questions that encourage a particular response
  - Example: "Do you support the president's use of targeted killing by remotely controlled drone aircraft to promote the safety and security of all Americans?" !!!!!
  - Example: "Do you agree with the recent Supreme Court decision…"
    - If asking such a question, you might want to ask the respondent how familiar they are with the S. Ct, and give them the context and holding of the decision then ask a shorter and less biased question

SUBTLE WORDING AFFECTS CONCLUSIONS DRAWN

More support

- "Assistance to the poor"
- "Affirmative action based on income"
- "Halting the rising crime rate"
- "Dealing with drug addiction"

- Relatedly – "imaginative prediction" easy to visualize
  - Diabetes
  - Car accident

Less support

- "Welfare"
- "AA based on race"
- "Law enforcement"
- "Drug rehabilitation"

- Harder to visualize
  - Homicide
  - Stomach cancer

ATTITUDES AND BEHAVIORS HAVE LIFE CONSEQUENCES

POLITICS • THE D.C. BRIEF

Ohio Voters Favor Abortion Access Yet Might Vote Against It

4 MINUTE READ

- 59% of midterm voters said abortion should generally be legal
- The Ohio Ballot Board chose to summarize the bill using phrasing that is incomplete in the eyes of pro-abortion rights activists, and describes the protections as associated with an “unborn child” rather than a “fetus.”
- Last month, pollsters at Ohio Northern University tested two versions of a question about the ballot measure. In one, they used the “unborn child” wording that is on ballots, and it drew 52% approval. But when phrased in the language of a “fetus” approval jumped to 68%.
- The group behind the amendment, Ohioans United for Reproductive Rights, challenged the ballot language in the Ohio Supreme Court. The court found the reference “citizens of the State of Ohio” to be misleading and ordered the change, but allowed the other modifications to remain.

https://www.nbc4i.com/news/your-local-election-hq/how-changed-ballot-wording-could-influence-support-for-ohios-issue-1-abortion-amendment/

PSEUDO-OPINIONS AND CONTEXTS

Table 1

Responses to "For each of the following, are immigrants from Latin America making [the situation] in Illinois better, worse, or not having much effect?" by Race

Group	White			Black		
Situation	Better	Not Much Effect	Worse	Better	Not Much Effect	Worse
Food, music and the arts	66%	29%	5%	57%	37%	6%
Economy	32	29	40	19	47	35
Quality of life	32	43	25	19	65	16
Social and moral values	35	42	22	13	71	16
Politics and government	26	41	33	19	63	18
Quality of public schools	16	33	52	19	64	17
Opportunities for you and your family	7	62	32	8	49	43
Taxes	12	42	47	5	43	52
Crime	6	28	67	4	28	68

Source: Data in Tables 1 and 2 are from the 2003 Chicago-Area Survey. Discrepancies are due to rounding.

sticity

.n"

Barboza, G., Knight, R. A., & Ready, T. (2007). *What do black and white residents of metropolitan Chicago think about Latin American and Mexican immigrants?: findings from the Chicago-Area survey*. University of Notre Dame, Institute for Latino Studies.

CLARITY

- Seems obvious, it's not
- Researchers know the topic and may be confusing or, they may know very little and fail to be specific enough
  - Example: "What do you think about the Governor's decision concerning prison furloughs?"
  - Example: "The DOC should stop releasing inmates on weekend furloughs and concentrate on rehabilitation"
- Short questions are best
- Avoid negative items

QUESTION ORDER, PRIMACY AND RECENCY EFFECTS

- Answers can also be influenced by the order in which response alternatives are presented.
- One of the most common response-order effects is a recency effect, in which the same answer is chosen more frequently when it appears as the last alternative in a series
- Examples:
  - "Should divorce in this country be easier to obtain, more difficult to obtain, or stay as it is now?"
  - "Should divorce in this country be easier to obtain, stay as it is now, or be more difficult to obtain?"
- First impression counts
  - People assume the first (or last) is most important and are more likely to remember it in different situations

OPEN V CLOSE ENDED QUESTIONS

- Potential to get differences in responses
- Sometimes useful to conduct a pilot study using open-ended questions before developing close ended questions
- Ex. How well off are you?

Fewer People Mention Economy in Open-Ended Version

What one issue mattered most to you in deciding how you voted for president?

	Open-ended[1]	Closed-ended[2]
The economy	**35%**	**58%**
The war in Iraq	5	10
Health care	4	8
Terrorism	6	8
Energy policy	*	6
Other	43	8
Candidate mentions	9	-
Moral values/social issues	7	-
Taxes/dist. of income	7	-
Other issues	5	-
Other political mentions	3	-
Change	3	-
Other	9	-
Don't know	7	2
	100	100

Data from Pew Research November 2008 Post-election survey

[1] Unprompted first responce to open-ended question.

[2] First choice from five options read to respondents.

SUMMER YOUTH SURVEY EVALUATION

- 88% claimed their family was ‘very’ or ‘fairly’ well-off
- But what does it mean to be well-off?
  - Having basics
    - Living, being alive, surviving, having needs met, comfortable, OK, not living in a trashy area
  - Stable, safe and stress free
    - Maintaining, problem-free, stability, peaceful, stress-free, supported,
  - Thriving
    - “fortunate,” “great,” “income,” “middle class,” and “rich” independent

OPEN AND CLOSE ENDED QUESTIONS

- Two options to ask questions
  - "What is the most important crime problem facing the police in your city?"
    - Fill in the blank
    - Given a list of options to choose
  - Pros and cons
    - Don't' prompt with open ended but cannot easily interpret the "fragments" and they are not conducive to data analysis, you must code them yourself
    - Ex. "burglary" "stealing stuff" – are these the same?
    - However close ended responses MUST be exhaustive and mutually exclusive

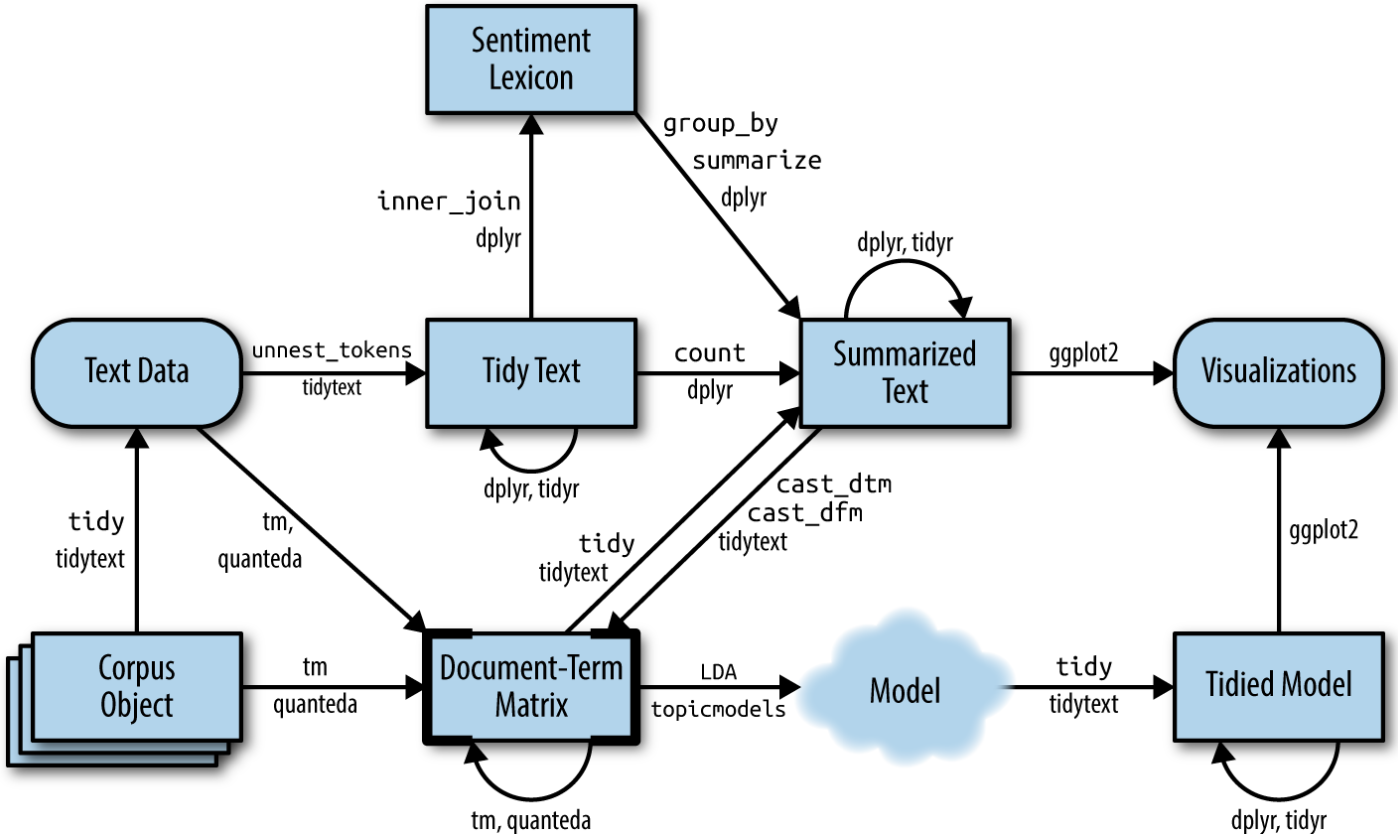
QUICKLY ANALYZING SURVEY DATA WITH JAMOVI

WEB SCRAPING

- Example: Mining Child Homicides

ANALYZING SURVEY TEXT DATA USING NATURAL LANGUAGE PROCESSING IN R

- A lot of open-ended response data goes totally ignored
- This can be a rich source of data and 'easily' analyzed in R



https://rpubs.com/guansaw/923951

Mining Information from Fatality and Near Fatality Reports

Incident Information:

Date of Incident: 10/27/14 **Date CP&P Notified of Incident**:10/14/14

Allegation Type: ☐ Abuse ☒ Neglect

Disposition: ☒ Substantiated ☐ Suspected/Pending

Description of Incident: On 10/27/14, G.G. was co-sleeping with her father, Ryan Geary. Whe he woke, Mr. Geary discovered that G.G. was unresponsive. G.G. was transported to the hospital; where she was pronounced at 12:15 a.m. While caring for his daughter, Mr. Geary admitted to taking illegal and prescription drugs. CP&P has completed a Child Protective Service investigation and substantiated Mr. Geary for neglect/child death.

Other Possible Child Victims and Living Arrangement (Gender/age of siblings or other children residing with the child victim or perpetrator only): None

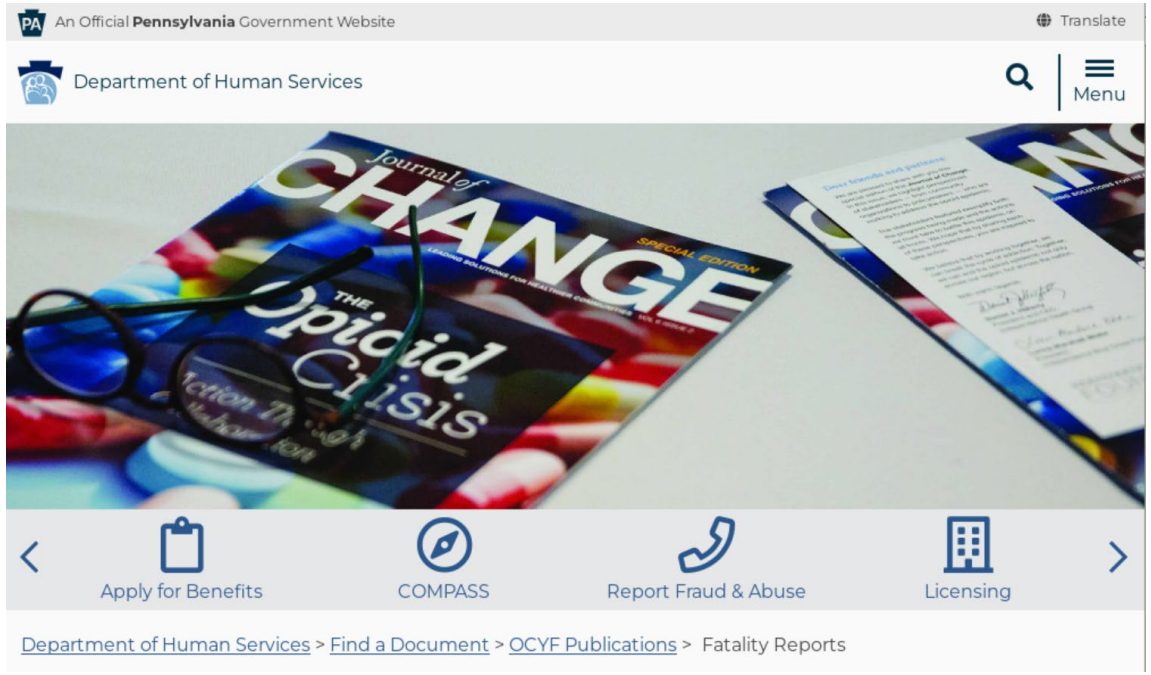
Historical Information:

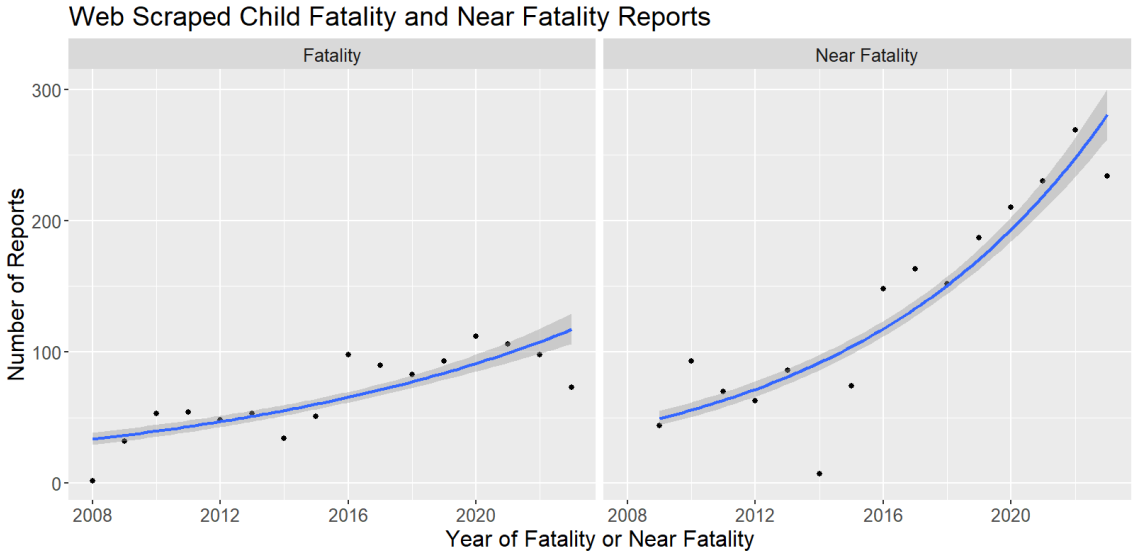
CP&P's Involvement with the Child Victim or Family PRIOR to Fatality or Near Fatality:

☐ Open (Date Opened:) ☐ Closed (Date Closed:) ☐ Not Previously Known

☒ The victim or family was known to CP&P but the details of such involvement are not pertinent to the fatality or near fatality and not subject to disclosure.

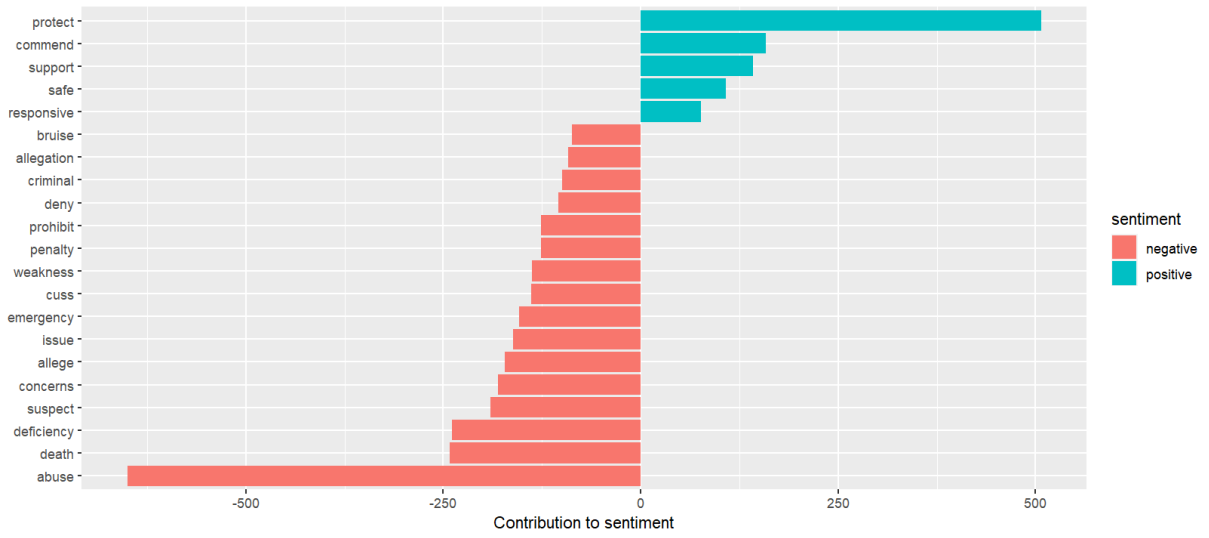
USING NLP TO MAKE SENSE OF MASSIVE TEXT DATA



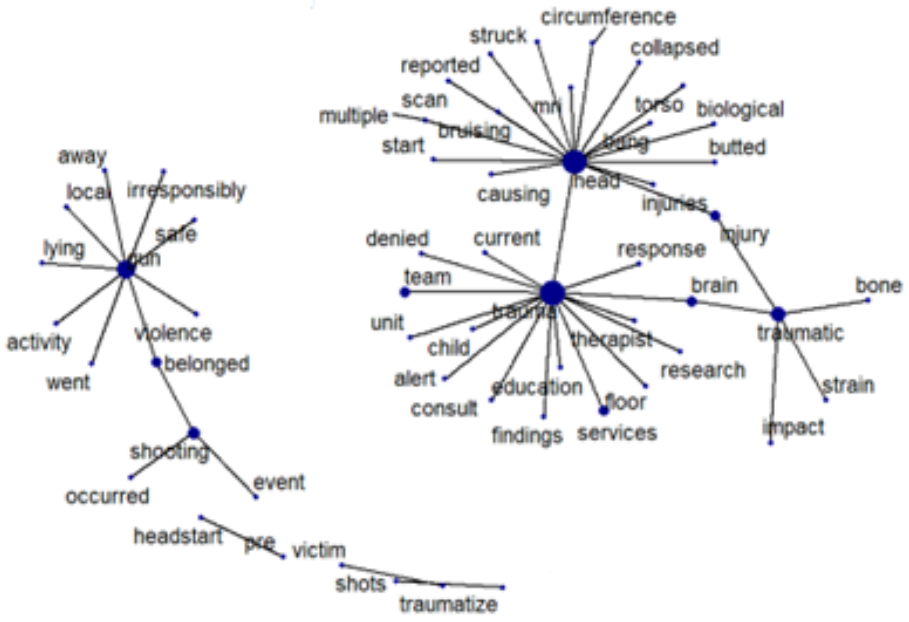


Run fatal-near-fatal.R

Sentiment Lexicon



Tidied Model Visualizations

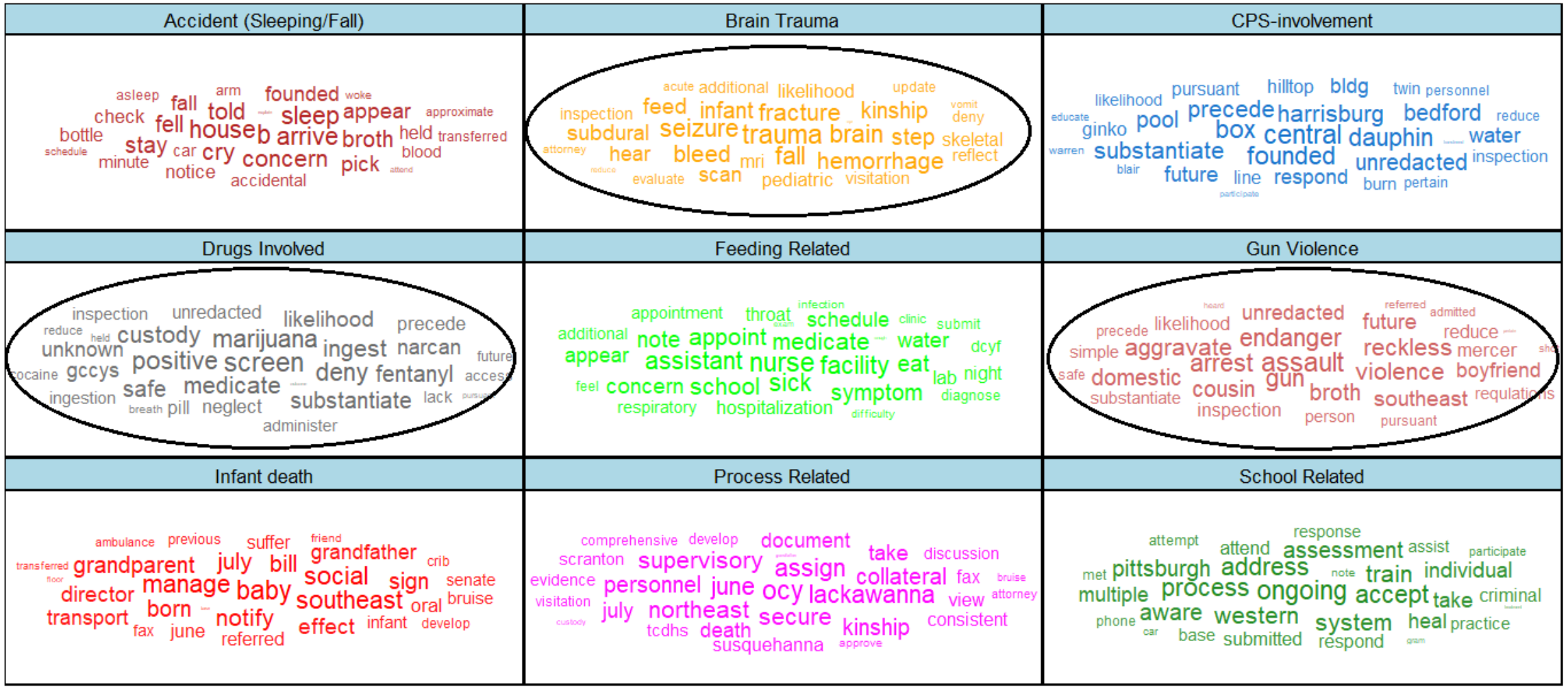


NLP FATALITY REPORTS

- R packages
- Descriptive Analysis
- Topic Models

DTM Summarized Text

Topics Emerging from Text Mining over 3,000 Fatality and Near Fatality Reports



Tidied Model → Visualizations

QUESTIONNAIRE CONSTRUCTION

- How a survey is constructed depends on how it will be administered
  - Self-administered
  - In person interview
  - Telephone interview

GENERAL QUESTION FORMAT

- Should be uncluttered
- Use of contingency questions – not all questions will pertain to all respondents
  - Batteries of questions about victimization will only be relevant to crime *victims*
- Matrix Questions – same set of answer categories used by multiple questions

EXAMPLES

1. Rape is having forced sexual intercourse that is against one's will. Have you ever been the victim of a rape?

- Yes [go to Q. 2]
- No [skip to Q 8]

2. Who were you raped by?

- An acquaintance [go to Q. 3]
- A stranger [skip to Q. 8]

8. Please indicate whether you strongly agree (SA), agree, disagree or strongly disagree (SD) with the following questions

	SA	Agree	Disagree	SD
a. Rape is a big problem in today's society				
b. Police in America shouldn't carry guns				
c. Schools should be gun free zones				

AGAIN ORDER MATTERS

- The content of one question can affect the answers given to prior questions
- In this situation, it may be preferable to ask the second question first
- In any event, what are the advantages and disadvantages of the matrix format?

DON'T REINVENT THE WHEEL

- There are tons of surveys out there, each is accompanied by a code book
- There are a lot of validated scales you can use

SURVEY ADMINISTRATION

- Choose appropriate modes (e.g., face-to-face, online).
- Consider logistics, timing, and ethical considerations.

DATA ANALYSIS

1. Discuss quantitative analysis techniques (e.g., descriptive statistics, correlations).
2. Consider inferential methods (e.g., t-tests, ANOVA) for hypothesis testing.